MILITARY SPECIFICATION

SEARCHLIGHTS, INCANDESCENT LAMP AND

MERCURY-XENON VAPOR LAMP, SIGNALING, 12-INCH

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 12-inch signaling searchlights.

1.2 Classification. The searchlights shall be of the following types as specified (see 6.2.1):

Type I - Incandescent
Type II - Mercury-Xenon vapor

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements for.
TT-E-490 - Enamel, Silicone Alkyd Copolymer, Semigloss (For Exterior and Interior Use).
TT-P-645 - Primer, Paint, Zinc Chromate, Alkyd Type.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915 - Cable and Cord, Electrical; For Shipboard Use, General Specifications for.
MIL-G-2697 - Glasses, Portlight, Circular, Heat Treated.
MIL-R-2728 - Reflector, Light, Parabolic, Searchlight.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/9 - Plates, Identification.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-P-15328 - Primer (Wash), Pretreatment, Blue (Formula No. 117-B for Metals).
MIL-P-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of.
MIL-L-18052 - Lamp, Mercury Xenon Vapor, Searchlight.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL
FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MS 15605 - Lamp, Incandescant, T-20, Mogul Bipost Clear Finish.
MS 25015 - Lamp, Incandescant, T-20 Bulb, Mogul Bipost Base.

DRAWINGS
NAVSHIP 18-T-4199-1 - Hood for 12-inch Signaling Searchlight.
NAVSHIP 815-1197106 - 12-Inch Mercury Xenon Signaling Searchlight.
NAVSHIP 9500-S6600-8149 - 12-Inch Signaling Searchlight.
NAVSHIP 9500-S6600-8150 - 12-Inch Signaling Searchlight.
NAVSHIP 95123-S6600-691438 - 12-Inch Searchlight, Venetian Shutter Assembly.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION
Hazardous Materials Regulations, 49 CFR Parts 171-177.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production a sample shall be examined and tested as specified in 4.3 (see 6.5).

3.2 Material. Material shall be as specified herein and meet the specified performance requirements. Reclaimed materials shall be used to the maximum extent possible.

3.2.1 Nonmagnetic material. When specified (see 6.2.1), nonmagnetic material shall be substituted for magnetic material wherever practicable. The magnetic permeability shall be 2.0 or less after fabrication. When the contractor feels that this requirement is not feasible, he should give detailed reasons and request waiver of the requirement from the Naval Ship Engineering Center (NAVSEC) via the contracting officer (see 6.4).

3.2.2 Dissimilar metals. Contact of dissimilar metals shall be avoided. An exception is made in the case where one of the metals is steel.

3.2.3 Protection against corrosion.

3.2.3.1 General requirement. All parts of the searchlight shall be of corrosion-resistant material or of material protected against corrosion.

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3.2.3.1.1 Corrosion-resistant materials. Silver, corrosion-resistant steel, copper, brass, bronze, copper-nickel, and nickel-copper alloys are corrosion-resistant materials within the intent of 3.2.3.1. Corrosion-resistant materials are required for bolts, nuts, studs, pins, cap screws, and similar fasteners, and for parts that move with respect to each other (have common bearing surfaces) in normal usage of the equipment.

3.2.3.1.2 Corrosion-resistant treatment. Electrodeposited zinc in accordance with type II, class 2 of QQ-Z-325 shall be used to protect those materials which are not corrosion-resistant.

3.2.3.2 Painting of searchlight equipment. Prior to painting, all external metal surfaces shall be given two coats of zinc-chromate primer conforming to TT-P-645. The first priming coat shall be applied after machine work has been completed and after surfaces to be covered have been thoroughly cleaned and dried and given a corrosion-resistant treatment, if applicable. Priming coats shall be followed by two coats of paint conforming to the following requirements:

(a) Painting of external surfaces shall conform to the color, gloss, flexibility, resistance to water immersion, and reflectance of Fed. Std. 595 color number 26270 of TT-E-490. The paint shall not discolor nor become soft when the searchlight is operated at rated voltage for a period of 2 hours, with the shutter closed.
(b) Painting of internal surfaces shall conform to the following requirements:
    (1) Paint shall be dull black.
    (2) Gloss shall be not greater than 4 units when determined by method 6101 of FED-STD-141.
    (3) Reflectance shall be not greater than 10 when determined by method 6121 of FED-STD-141.
    (4) Under normal operating conditions, the paint shall not fume to such an extent as to be harmful to personnel or cause an objectionable film to form on the interior of the searchlight.
    (5) Flexibility and resistance to water immersion shall conform to (a).

3.2.3.2.1 Alternate painting requirements. Alternate painting requirements shall be in accordance with the following:

(a) Primer in accordance with MIL-P-15328.
(b) External surfaces in accordance with MIL-E-15090.

3.3 Design and construction. Searchlights shall be designed and constructed in conformance with Drawings 9500-S6600-8149, 9500-S6600-8150, 815-1197100 and 18-T-4199-L as specified. Design and construction shall provide for continued satisfactory operation and will not require special tools or excessively precise workmanship to perform routine maintenance. Material and construction shall be the lightest and most compact consistent with the strength required for safety and reliability. Special tools are defined as those tools not listed in the National Supply Catalog. (Copies of this Catalog may be consulted in the office of the Defense Contract Administration Service (DCAS).)

3.3.1 General design. The searchlight shall be designed for long service, without frequent overhaul, under severe conditions of shock, gun blast, vibration, humidity, salt atmosphere, and the rolling and pitching of a ship. The searchlight shall be light tight when the lamp is burning (no light shall be visible at any time from any part of the searchlight from a distance of 100 feet with the front door completely covered). Design and construction of the searchlight and the signaling shutter shall be such that messages transmitted with the searchlight at speeds up to 15 words (300 operating cycles of the shutter) per minute can be read by an experienced signalman.

3.3.2 Interchangeability. Similar parts, including repair parts, for equipment furnished on the same order or built to the same manufacturer's drawings shall be strictly interchangeable without the necessity of further machining or handfitting of any kind.

3.3.3 Threaded parts.

3.3.3.1 Screw threads. Screw threads shall conform to the requirements of Handbook H28.

3.3.3.2 Clamping devices. Clamping devices shall be secured in such a manner as to prevent loss during operation and maintenance.

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3.3.4 Component parts. Component parts of the searchlight shall consist essentially of a drum with a rear door signaling shutter, lamp and accessories, reflector, front door glass, yoke, supply cable, hood with filters, and those parts specified in 3.3.4.1 through 3.3.7.1.

3.3.4.1 Drum.

3.3.4.1.1 Mounting. The drum shall be mounted on trunnions which shall seat on the yoke. Movement of the drum in the vertical plane shall be as specified in 3.5.11. There shall be a single clamp for locking the searchlight in any angle of elevation.

3.3.4.1.2 Mounting brackets. Two mounting brackets for infrared viewer shall be installed. These brackets shall conform to figure 1. Mounting brackets shall be of corrosion-resistant steel.

3.3.4.1.3 Sights. Two sights, each of which shall consist of a front and rear piece, shall be mounted, one on each side of the drum near the top for aiming the beam. The rear piece shall be a globe ring with a 3/4-inch diameter opening. The front piece shall have a small ball or a pointed top. The two pieces shall be mounted as far from each other as practicable and parallel to the axis of the searchlight beam.

3.3.4.2 Rear door. The rear door shall be hinged to provide access to the inside of the drum, and shall be clamped to the drum. The fit of the rear door to the drum shall be such that the location of the reflector with respect to the filament is positively fixed when the door is closed and clamped in position. The door shall include provision for easy mounting of the reflector. A handle, designed to permit aiming of the light by a man wearing heavy gloves, shall be attached to the rear door. The handle shall be conveniently placed for continuous training and elevating of the drum while signaling. This handle shall be sufficiently heat insulated from the drum to prevent its temperature exceeding 55°C during continuous signaling operation for 4 hours, assuming an ambient temperature of 25°C.

3.3.4.3 Signaling shutter.

3.3.4.3.1 General design. The signaling shutter shall be of the lattice vane ("venetian blind") type and shall be in accordance with Drawings 9500-S6600-8149, 9500-S6600-8150, 815-1197106, and 95123-S6600-691438. The light leakage through the signal shutter when closed shall be reduced to a minimum, and such leakage, if present, shall not cause a reduction in signal efficiency. The signaling shutter assembly (with the front door glass removed) shall reduce the average maximum beam candlepower not more than 20 percent.

3.3.4.3.2 Shutter frame. The shutter frame which serves to hold the shutter vanes and front door glass shall be of the fixed type, and shall be held rigidly to the drum by hexagonal headed bolts. Bolts shall be spaced at 45 degrees from each other on a 13-1/4-inch bolt circle diameter. Two of the holes shall be located at the ends of the vertical diameter. The number, size, and location of these bolts shall be in accordance with Drawing 9500-S6600-8150.

3.3.4.3.3 Interchangeability. Special care shall be exercised in following the drawings referred to in 3.3.4.3.1 with regard to the features involving fit of the assembled shutter either on the front of the drums being supplied, or on other searchlights designed to accommodate the standard shutter. Drawing 9500-S6600-8149 is modified to the extent that the cut in the drum to accommodate the shutter boss is not required.

3.3.4.3.4 Construction. Design and construction of the shutter shall be such that it shall be capable of withstanding 4,000,000 cycles of opening and closing without wear or damage to such an extent as to preclude continuation of satisfactory operation. Two signal shutter operating handles shall be provided, one on each side of the drum. The two handles shall be completely interchangeable.

3.3.4.3.5 Adjustable stops shall be provided with both handles to avoid twisting the drive shaft in opening and closing the shutter. Two counter springs (preferably interchangeable) shall be provided for returning the shutter vanes to the fully closed position. One spring shall be placed on each side of the shutter and shall be so designed that the springs may be easily replaced. The throw of the shutter operating handle for movement of the shutter vanes, from fully closed to fully open, shall not exceed 90 degrees. A positive locking device shall be provided for locking the shutter in the open position. The lock shall be so designed that there shall be no possibility of interference with the shutter operation. Provision shall be made for easy mounting of the front door glass, which shall be firmly held in place. Design of the glass holding frame shall be such that the securing pressure can be maintained in service.

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3.3.4.4 Lamp and accessories (type I).

3.3.4.4.1 Lamp. A 30 volt or 115 volt, 1,000 watt mogul bipost lamp as specified in MS 25015-1 or MS 15605 shall be supplied with each searchlight. The lamp voltage to be supplied shall be as specified (see 6.2.1).

3.3.4.4.2 Lamp socket. The lamp socket shall accommodate the bipost lamp, and shall be provided with one floating clamp. Means shall be provided to ensure rapid drainage of water from all surfaces and the bipost pin receptacles. Maximum creepage distances between live parts of the socket shall be provided.

3.3.4.4.3 Lamp socket mounting. The socket and its mounting boss on the inside of the drum shall be positioned so that the filament of the incandescent lamp shall be centered at the focal point of the reflector. In order to provide for slight inaccuracies in the reflector position, the socket mounting holes shall be slotted to permit adjustment of focus at the time of assembly. The size of the slots shall be adequate to provide for the range of adjustment required due to variations in lamp, reflector, and searchlight.

3.3.4.5 Lamp and accessories (type II).

3.3.4.5.1 Lamp. The type II lamp to be supplied with the searchlight shall be in accordance with MIL-L-18052.

3.3.4.5.2 Starter assembly. The starter assembly shall be mounted on the searchlight in conformance with Drawing 815-1197106. This assembly shall contain the required circuitry for igniting the lamp. The starter assembly shall be removable for maintenance.

3.3.4.5.3 Control assembly. The control assembly shall be mounted separate from the searchlight in conformance with Drawing 815-1197106. The control assembly shall pass applicable tests required by the searchlight.

3.3.4.6 Reflector.

3.3.4.6.1 Material. Reflectors shall conform to MIL-R-2728.

3.3.4.6.2 Mounting. Reflectors shall be so mounted as to allow for contraction and expansion and to prevent injury resulting from ship's vibration and shock of gunfire. The reflector shall not shift position when the searchlight is elevated or depressed, and shall not be subjected to distorting stresses. Temperature variations encountered during operation shall not cause distorting stresses. The reflector shall be positively centered and the holding pressure shall be maintained during the specified service conditions.

3.3.4.7 Front door glass. The front door glass shall be a flat heat-treated glass 12-1/2 inches in diameter by 1/4 inch thick. The glass shall conform to type A of MIL-G-2697.

3.3.4.8 Yoke and socket.

3.3.4.8.1 Yoke. A vertical pedestal rod shall be provided at the base of the yoke so that the whole searchlight may be rotated in azimuth upon the release of a lock or clamp, and so that the light may be held at any angle of train through 360 degrees. The mounting of the trunnions in the yoke and the pedestal rod in its socket shall be such that the whole shall be easily demountable.

3.3.4.8.2 Socket. The pedestal rod shall seat in a socket with such shape and strength that it can be bolted or riveted to the plating of the ship's structure or bolted to a vertical rod between 3 and 3-3/4 inches in diameter.

3.3.4.9 Cable. A length of 15 feet of cable in accordance with the requirements for type THOF-9 of MIL-C-915 shall be supplied with each searchlight. The dimensions shall be measured outside the drum.

3.3.4.10 Hood assembly. The hood assembly, including lenses required for each type of searchlight, shall be in accordance with Drawing 18-T-4199-L.

3.3.4.11 Gaskets. Gasket material, where used (front door glass and reflector mountings, and similar items), shall have the following properties:

(a) Shall be adequately pliable and of sufficient thickness to provide necessary cushioning effect.
(b) Shall not absorb moisture.

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(c) Shall be able to withstand the temperature variations (minus 40°C to plus 170° C) expected in service without charring, or volatilizing.
(d) Shall not separate, shrink, or become hard and brittle in service.
(e) Shall retain its shape during service.
(f) Shall be of asbestos - free material.

3.3.5 Drainage. Drainage holes shall be placed in both the shutter housing and drum for draining any moisture which may accumulate in either the drum or shutter housing. Drainage holes shall be placed in the lowest portion of the drum and shutter housing when the searchlight is held in a horizontal position and shall be so constructed that the amount of light leakage due to these holes shall be reduced to a minimum (see 3.3.1). Provisions for drainage shall be such as to prevent water from reaching any current-carrying parts in the drum under surface running conditions.

3.3.6 Weight. The weight of the searchlight shall be as follows:

(a) Type I - Exclusive of hood and repair parts, the weight shall not exceed 90 pounds. The weight of the hood, including aperture disc, filter and filter cover shall not exceed 6 pounds.
(b) Type II - Exclusive of hood and repair parts but including electrical controls, the weight shall not exceed 150 pounds. The weight of hood, including aperture disc, filter, and filter cover, shall not exceed 6 pounds.

3.3.7 Repair parts. Repair parts shall be interchangeable with parts installed on the original searchlight.

3.3.7.1 Onboard repair parts shall be furnished in accordance with MIL-P-15137 (see 6.2.1).

3.4 Technical data. The contractor shall prepare the following technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2), and as specified in 3.4.1 and 3.4.2.

3.4.1 Drawings. Drawings shall be in accordance with the data ordering document (see 6.2.2) and shall contain the following:

(a) A list of repair parts, the name of each item, the quantity of each item being furnished in each set of repair parts and the manufacturer's ordering identification. This list shall be a duplicate of the list which is placed inside the repair parts box.
(b) Table of weights:
  (1) Weight of searchlight assembly, complete.
  (2) Weight of repair parts, including boxes.
  (3) Weight of searchlight, crated with repair parts box.
(c) Dimensioned assembly outline (to scale) showing all principal dimensions in at least a front view and a side view.
(d) Mounting plate or foot with detailed dimensions for mounting.
(e) Painting instructions, including method of treatment of searchlight for painting, color, and applicable specification of paint.
(f) Data which shall appear on the searchlight identification plate to be supplied in accordance with the drawing.

3.4.2 Technical manuals. Technical manuals shall be furnished with searchlights and shall be in accordance with the data ordering document (see 6.2.2).

3.4.3 Identification and marking plates. Identification plates and other designating marking plates shall be made of corrosion-resisting steel or brass, in accordance with MIL-P-15024 and MIL-P-15024/5. These plates shall be of sufficient size, and shall be installed on and furnished as a part of the searchlight for which they are intended. They shall be attached to a part which shall not ordinarily be renewed during its service life, and shall be located in a readily accessible position. Data marked on the identification plate shall include the following items:

(a) Manufacturer's name.
(b) Identification symbols.
(c) Searchlight serial number.
(d) Contract number.
(e) Date of manufacture (year).
(f) "Twelve-inch signaling searchlight, Model_____" (as assigned).

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(g) The marking "NM" stamped or embossed on the identification plate to designate nonmagnetic material (when applicable).
(h) Blank space for DCAS official stamping.

3.5 Operational requirements. The searchlight shall ignite within 15 seconds after power is applied. Upon ignition of the lamp, the starter circuit shall automatically disconnect itself from the circuit. This includes both hot starting and cold starting. Hot starting is defined as starting a lamp which has been off for a period of 60 ± 30 seconds after having been continuously energized for at least 5 minutes. An off period exceeding 2 minutes constitutes a cold start.

3.5.1 Power requirements.

3.5.1.1 Type I. Type I searchlight shall operate at either 30 volts or 115 volts, direct current or alternating current (see 3.3.4.4).

3.5.1.2 Type II. Type II searchlight shall operate at the following power requirements:

(a) Input voltage - 115 volts.
(b) Frequency - 60 hertz (Hz).
(c) Starting current - 45 amperes.
(d) Operating current - 18 amperes.

3.5.2 Waterproofness. The searchlight shall be such that no water that might interfere with the operation of the searchlight shall enter the drum when subjected to the test of 4.6.9.

3.5.3 Safety. The searchlight shall not expose operating personnel to hazards from inadvertent contact with electric potentials and with high temperatures.

3.5.3.1 Electric shock. Means shall be provided for automatically deenergizing the starting circuit when the searchlight door is open for relamping.

3.5.3.2 High temperatures. Where surface temperatures could reach temperature levels high enough to cause personnel injury, such surfaces shall be insulated to maintain surface temperatures below 55°C in an ambient temperature of 25°C.

3.5.4 Ambient temperature. Searchlights shall operate in an ambient temperature ranging from minus 30°C to plus 50°C.

3.5.5 Dielectric strength. The searchlight shall be capable of withstanding for 1 minute, a 60 Hz alternating current (a.c.) voltage with effective potential of 1240 volts applied between each conductor and searchlight frame.

3.5.6 Insulation resistance. The insulation resistance, with 500 volts applied between each conductor and the searchlight frame, shall be not less than 1 megohm and shall be measured before and after each dielectric strength test.

3.5.7 Shock. The searchlight and control assembly (for type II), less lamp, shall withstand high impact shock as specified in MIL-S-901 without fracture or deformation when tested as specified in 4.6.11.1.

3.5.8 Vibration. The searchlight shall operate during vibration tests specified in 4.6.11.2. Fasteners (bolts, screws, screw clamps, rivets, etc.) shall remain tight and rigidly fixed, and no damage shall be sustained by the assembly or its components.

3.5.9 Temperature limits. The temperature of the hottest part of the outside of the searchlight, less the cover glass, shall not exceed 130°C during continuous signaling operation for 2 hours, assuming an ambient temperature of 25°C. This temperature limit does not apply to the control unit of type II searchlights.

3.5.10 Illumination. The searchlight shall be tested as specified in 4.6.8 through 4.6.8.5.

3.5.11 Inclination and training. The searchlight shall provide for minimum movement of the drum in the vertical plane of 105 degrees above and 45 degrees below the horizontal, and for turning the searchlight through a horizontal angle of 380 degrees.

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3.5.12 Salt spray. The searchlight shall withstand prolonged exposure to salt spray without damaging effect to the unit (see 4.6.10).

3.5.13 Temperature-humidity. The searchlight unit shall be capable of operation at 95 percent humidity and at temperatures ranging from minus 25°C to plus 50°C (see 4.6.5.2.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in 4.5 and 4.6.

4.3.1 First article inspection report. For each procurement, the contractor shall submit to the procuring activity a complete report of the first article inspection conducted. Also, one copy of this report shall be forwarded to the following activities:

(a) NAVSEC.
(b) Cognizant DCAS.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination specified in 4.5, and the tests of 4.6.1 through 4.6.6, and 4.6.16.

4.4.1 Sampling.

4.4.1.1 Lot. For purposes of sampling, a lot shall consist of not more than 100 searchlights offered for delivery at one time.

4.4.1.2 Sampling for quality conformance inspection. A random sample of searchlights shall be selected from each lot and shall be subjected to each of the inspections specified in 4.4 with sampling and quality conformance in accordance with inspection level III and AQL 1.5 (normal inspection) of MIL-STD-105.

4.5 General examination.

4.5.1 Visual and dimensional examination. Sample searchlight shall be visually and dimensionally examined to determine conformance with this specification and applicable drawings.

4.5.2 Adjustment, fit, and materials. Sample searchlights shall be subjected to a thorough examination to ascertain that the design, materials, workmanship and weight are in strict conformity with this specification. The fit of parts shall be observed with respect to the interchangeability of such parts as are likely to require replacement during normal service life of the equipment.

4.5.3 Interchangeability. Examination shall be made to verify compliance with the requirement of 3.5.2 and 3.5.4.3.3.

4.5.4 Repair parts. Repair parts of sample searchlights shall be subjected to a careful examination to ascertain that the materials, workmanship, and finish comply fully with the manufacturer's drawing.

4.5.5 Safety. Each sample searchlight shall be examined to determine conformance to 3.5.3.

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4.5.6 Inclination and training. Sample searchlights shall be examined to verify compliance with the requirements of 3.5.11.

4.6 Test procedures.

4.6.1 Focus. Searchlights shall be checked for compliance with 3.3.4.4.1 concerning focusing range and general conformance with the focusing provisions on Drawing 815-1197106.

4.6.2 Shutter adjustment (partial). Searchlights shall be tested to determine that the shutter is adjusted to cause minimum reduction in the average maximum beam candlepower when wide open, and to cause the minimum emission of light when closed. The searchlight shall also be checked to ensure that the shutter handle stops on each side of the shutter strike, the shutter stops simultaneously both when open and when closed.

4.6.3 Sighting test. Searchlights shall be positioned 57.3 feet from a target, with the lamp filament at the focus of the reflector, and burning, the line of sight from both sights shall fall within 2 feet of the point of maximum "beam candlepower".

4.6.4 Insulation resistance and dielectric strength tests. Insulation resistance and dielectric strength tests shall be made as follows to determine conformance with 3.5.5 and 3.5.6:

(a) Before any other tests.
(b) After waterproofness test.
(c) Twenty four hours after salt spray test.

These tests shall be applied both cold and after light has reached its maximum operating temperatures, when operated in an ambient air temperature of 25°C.

4.6.5 Operation.

4.6.5.1 Type I. The power input shall be measured when the type I searchlight is connected to the applicable sources specified in 3.5.1.1.

4.6.5.2 Type II. When connected to the power source specified in 3.5.1.2, the type II searchlight shall meet the operational requirements of 3.5.

4.6.5.2.1 For first article tests only. When connected to the power source specified in 3.5.1.2, a temperature-humidity test shall be conducted at 95 percent humidity. One cold start and three hot starts shall be made after a 12 hour soak at minus 25°C and repeated after a 12 hour soak at plus 50°C. A start exceeding 20 seconds shall be cause for rejection.

4.6.6 Light tightness. The light tightness of the searchlight with the lamp burning at rated voltage, shall be checked to determine compliance with 3.3.1 both with and without the shutter in operation. The test shall be made with the front door glass completely covered.

4.6.7 Shutter adjustment. The amount of stray and spill light from the searchlight shall be determined. For this test the shutter shall be attached but the front door removed, and the shutter adjusted to produce minimum reduction in average maximum beam candlepower.

4.6.8 Illumination.

4.6.8.1 Method of measurement.

(a) The searchlight shall be focused for minimum beam spread at a distance of 57.3 feet.
(b) The distance between the reflector and the target at which measurements are made shall be a minimum of 57 feet. A longer range from target to reflector of 100 to 150 feet is preferred, and shall be employed if available.
(c) The screen or target at which illumination measurements are made shall be marked with horizontal and vertical lines, so spaced that a minimum of 225 equally spaced intersections of horizontal and vertical lines fall within the beam spread. The foot-candle value at each intersection of the lines shall be measured. Candlepower values at each intersection shall be determined by multiplying the foot-candle reading at the intersection and the square of the distance in feet from the reflector to the target. Curves of candlepower values shall be plotted where required.

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(d) The average maximum beam candlepower of the searchlight shall be determined by averaging the 10 highest values of illumination in foot-candles measured at the intersections of the horizontal and vertical lines and multiplying this average value by the square of the distance from the reflector to the target. The beam spread shall be expressed in terms of the plane angles, horizontally, vertically, or otherwise considered, subtended at the reflector by the points at which the candlelight falls to 10 percent of the average maximum beam candlepower.

(e) In each case where the average maximum beam candlepower is determined, the maximum beam candlepower of the searchlight shall also be determined. This determination shall be made by searching the central area of the target for the maximum illumination.

(f) Type I searchlights. Illumination measurements shall be taken with an aged and calibrated incandescent lamp operating at or near rated voltage, so that its luminous intensity in the direction of the reflector shall be a maximum, and the same value for all conditions of test, and for similar tests of all searchlights of the same type.

(g) Type II. Illumination measurements shall be taken with an aged and calibrated mercury-xenon lamp operating at or near rated wattage so that its luminous intensity in the direction of the reflector shall be a maximum and the same value for all conditions of test, and for similar tests of all searchlights of the same type.

4.6.8.2 Measurements required. Beam candlepower data, as necessary, shall be taken and recorded to determine compliance with the requirements of 3.3.4.3.1 concerning average maximum beam candlepower reduction caused by the shutter. This shall include the average maximum beam candlepower under the following conditions:

(a) With the shutter removed from the searchlight.
(b) With the shutter in place, but the front door glass removed as follows:
  (1) With the shutter as received from the manufacture.
  (2) With the shutter adjusted to produce minimum reduction in average maximum beam candlepower.
  (3) After 500,000 cycles of operation.
  (4) After 1,000,000 cycles of operation.
  (5) After 2,000,000 cycles of operation.
(c) With the searchlight completely assembled.

4.6.8.3 Beam pattern. Complete candlepower curves as specified in 4.6.8.2 shall be taken for the following conditions:

(a) With the shutter removed.
(b) With the shutter attached but front door glass removed, and shutter adjusted.
(c) With searchlight completely assembled.

4.6.8.4 Stray and spill light. The amount and source of stray and spill light from the searchlight shall be determined.

4.6.8.5 In determining the reduction in average maximum beam candlepower caused by the shutter, only the following values shall be considered; other illumination measurements shall be taken for information only, and shall have no bearing on the acceptance of the model, unless the manufacturer has departed from the specified design:

(a) Average maximum beam candlepower with the shutter removed from the searchlight.
(b) Average maximum beam candlepower with the shutter attached, but front door glass removed, and with the shutter adjusted to produce minimum reduction in average maximum beam candlepower.

4.6.9 Waterproofness. Compliance with 3.5.2 shall be checked by playing a stream of water from a hose 1 inch in diameter under a head of approximately 35 feet from a distance of 10 feet on the searchlight for 5 minutes. The stream shall be projected against the searchlight from all directions, including the bottom. During this test the searchlight shall be operated in the horizontal position.

4.6.10 Salt spray test. The searchlight unit complete with signaling shutter, and with flange of hood attached; and with lamp of type I searchlight burning at rated voltage in cycles of operation consisting of 6 minutes on and 6 minutes off, shall be placed in a salt spray machine, and subjected to the following test. For type II searchlight, the lamp shall not be installed:

(a) The searchlight shall be subjected to the action of a salt spray for 200

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hours. This salt spray shall consist of a spray of a 20 percent solution of commercially pure sodium chloride in distilled water. The spray shall be at 55°C and shall be applied alternately with a 3-minute drying period of air at 55°C producing a 3-minute wet and 3-minute dry condition continuously under ultra-violet light. Upon completion of the test, the searchlight shall be washed with fresh water and examined.

(b) Careful examination after the salt spray test shall show no damaging effect to the unit due to the salt spray. In addition, the searchlight shall function satisfactorily in accordance with this specification after the test. Lockwashers shall be thoroughly examined.

(c) The test specified in 4.6.10 (a) shall be conducted on the hood, less its mounting flange. Examination after this test shall show no damaging effect to the hood as a result of this test. It shall be possible to assemble and disassemble all parts of the hood.

4.6.11 Shock and vibration.

4.6.11.1 Shock.

(a) The searchlight shall be tested for class H.I. (high impact) shockproofness in accordance with MIL-S-901. The classification of the shock test shall be type "A" for the lightweight equipment. Damage to the lamp during the test shall not be cause for rejection.

(b) The searchlight shall be mounted by clamping to a vertical pipe or by bolting to a vertical plate. The following shall apply:

(1) Type I searchlight - The lamp shall be installed and burned at rated voltage.

(2) Type II searchlight - The lamp shall not be installed and the control assembly shall be tested.

4.6.11.2 Vibration. Searchlights shall be subjected to the type I vibration test of MIL-STD-167-1.

4.6.12 Gaskets. Gaskets shall be tested for compliance with 3.3.4.11.

4.6.13 Life test. The life test shall include operation for 4,000,000 cycles of the shutter at a rate of 200 cycles per minute. (A cycle consists of the movement of the shutter from the completely closed to the completely open positions and return to the completely closed position.) The shutter shall be attached to the drum for this test, and the lamp shall be burned at rated voltage throughout the duration of the test. Bearings of the shutter shall be lubricated at intervals of 250,000 cycles. The test shall be divided into two parts as follows:

(a) 2,000,000 cycles of operation of the shutter shall be performed prior to conducting the salt spray test specified in 4.6.10.

(b) 2,000,000 cycles of operation of the shutter shall be performed after conducting the salt spray test specified in 4.6.10 and the shock test specified in 4.6.11. Care shall be taken to remove salt incrustation after the salt spray test, and to lubricate the shutter bearings prior to beginning the shutter operation test.

The torque in inch-pounds shall be recorded at the beginning and end of each half of the operating test. Tests shall also be made to determine that messages transmitted with the searchlight at speeds up to 300 operating cycles per minute can be read by an experienced signalman. For this purpose, the variation in light output caused by the shutter shall be measured by means of an oscillograph or other suitable measuring device. This determination shall be made both prior to, and upon completion of the shutter operation test.

4.6.14 Temperatures.

4.6.14.1 Temperatures shall be measured with thermocouples at points outside of the searchlight to determine compliance with 3.5.3.2 and 3.5.9. The shutter shall be operated at 200 cycles per minute and the lamp shall be operated at rated voltage.

4.6.15 Shutter interchangeability. Interchangeability of the shutter with shutters specified on Drawing 95123-S6600-691438 shall be determined.

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4.6.16 Reflector mounting. To check for compliance with 3.3.4.6.2, the following test shall be made:

(a) A chart of photometric readings with the reflector in place but with the reflector clamps only "finger tight" shall be made. A similar chart of readings shall be made after the clamps are wrench tightened and the reflector brought to operating temperature. No change in beam axis shall be evident.

4.7 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation-packaging and packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation-packaging, packing and marking. Searchlights and repair parts thereto shall be preserved-packaged level A or C, packed level A, B, or C as specified (see 6.2.1) and marked in accordance with MIL-E-17555 and as specified herein.

5.1.1 Items containing metallic mercury or mercury compounds (see 3.3.4.5). The preservation-packaging and packing requirements specified in 5.1 for the level required shall comply with the requirements of the Department of Transportation (DOT), Materials Transportation Bureau, Hazardous Materials Regulations, 49CFR Parts 171-177.

5.1.2 Cushioning, dunnage and wrapping materials.

5.1.2.1 Level A preservation-packaging, level A and B packing. Use of all types of loose-fill materials for applications such as cushioning, filler, stuffing and dunnage for material destined for shipboard use is prohibited.

5.1.2.2 Level C preservation-packaging and packing. Unless otherwise specified in the contract or order, use of all types of loose-fill materials for applications such as cushioning, filler, stuffing and dunnage for material destined for shipboard use is prohibited. When approved for use by the contract or order, unit packages and containers (interior and exterior) shall be marked or labeled as follows:

"CAUTION

Contents cushioned etc. with loose-fill material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard, or transparent flexible cellular material."

5.1.3 Special marking.

5.1.3.1 Items containing metallic mercury or mercury compounds (see 3.3.4.5). Whenever an item containing mercury is enclosed in a container (interior package or shipping) which covers or hides the item identifying plate or label or when the item size causes identification to be impractical, each container shall be marked, tagged or labelled with the information stated herein. Tags and labels shall be affixed to the outer surface of each container and coated with a transparent non-fading waterproof coating. In addition, all packages and containers (unit, intermediate and shipping) shall be labelled with the DOT, class B poisonous material label."

CAUTION-MERCURY
USE APPROVED BY NAVSEA

Contains _____ Mercury or _____
mercury compound

in _____
Name/location of part

In case of spillage in excess of two ounces
by weight of mercury isolate area and notify
ship's Commanding Officer and Industrial Hygienist,
Medical Department or equivalent.

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6. NOTES

6.1 Intended use. Searchlights covered by this specification are intended for use as high-intensity signaling searchlights for shipboard use.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Whether nonmagnetic material is required (see 3.2.1).
(d) Lamp voltage required (see 3.3.4.4.2).
(e) Quantity of onboard repair parts required if different from 3.3.7.1.
(f) Preservation, packaging and packing or marking required (see 5.1).

6.2.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
3.4.1	Drawings, engineering and associated lists, level 3 (production)	DI-E-7015	Design activity designation - Contractor Drawing number - Contractor
3.4.2	Manual, technical, (type I), equipment	UDI-M-23994	Type I of MIL-M-15071
4.1.1	Inspection system program plan	DI-R-4803	------

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Departures from the general features of design specified in this specification should be fully described in bids.

6.4 The bidder shall include in the bid a list of nonmagnetic materials he proposes to furnish for the various searchlight components. The magnetic permeability shall be indicated for each material.

6.5 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.7 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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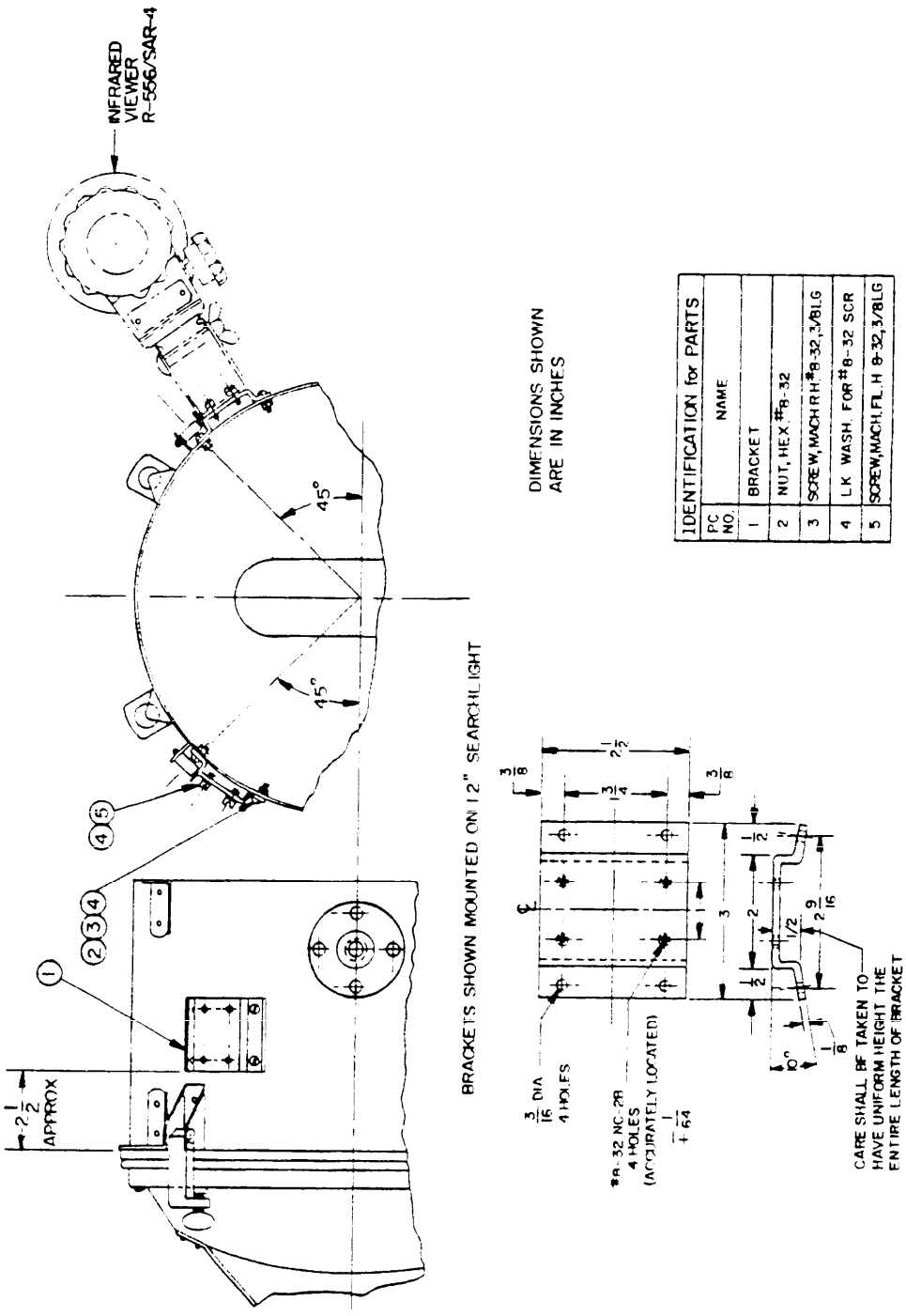


FIGURE 1. Mounting brackets.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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